Core summer and timing

X

P = (X + 6 VB) / 7

N = (LP + 2 BDAMP + 4 VB) / 7

CB = CL = 150 nF + 33 nF

BW

BDAMP

P

N

A

P2

N2

File: core_sum.kicad_sch

Integrator 1

P

N

O

P − N −> discrete current −> BO

File: xota1.kicad_sch

BP buffer

BO

BP

File: xbb.kicad_sch

Integrator 2

P2

N2

O

P2 − N2 −> discrete current −> LO

File: xota2.kicad_sch

LP buffer

LO

LP

File: xlb.kicad_sch

Resonance buffer

BW

BDAMP

Continuous BP attenuation

File: xres.kicad_sch

FULL DISCRETE VCF−1 / A0 DEVELOPMENT
P − N drives integrator 1, BP feeds integrator 2, LP and damped BP return to the summer.
Timing capacitors and the resonance control are on core_sum.
Input shape, HP reconstruction, Matter, MIX saturation, power, and bias are on the pages below.
No op−amp IC, OTA IC or reference IC. All electrical parts are THT.

power_cv

File: power_cv.kicad_sch

bias_1

File: bias_1.kicad_sch

bias_2

File: bias_2.kicad_sch

xvb

File: xvb.kicad_sch

floor_bias

File: floor_bias.kicad_sch

xv5

File: xv5.kicad_sch

command

File: command.kicad_sch

xvt

File: xvt.kicad_sch

current_servo

File: current_servo.kicad_sch

input_shape_1

File: input_shape_1.kicad_sch

input_shape_2

File: input_shape_2.kicad_sch

cv_buffer

File: cv_buffer.kicad_sch

xlb

File: xlb.kicad_sch

xgain

File: xgain.kicad_sch

xgb

File: xgb.kicad_sch

xas

File: xas.kicad_sch

hp_reconstruction

File: hp_reconstruction.kicad_sch

xhp

File: xhp.kicad_sch

matter

File: matter.kicad_sch

xalt

File: xalt.kicad_sch

xlh

File: xlh.kicad_sch

outputs

File: outputs.kicad_sch

xmb

File: xmb.kicad_sch

ports

File: ports.kicad_sch

test_points

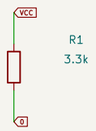
File: test_points.kicad_sch

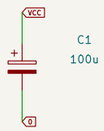
Sheet: /
File: mamut−vcf−1−a0−tht.kicad_sch

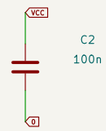
Title: Mamut VCF−1 / complete discrete A0

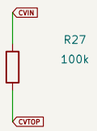
Size: A2 | Date: | **Rev: A0**
KiCad E.D.A. 10.0.7 | Id: 1/32



















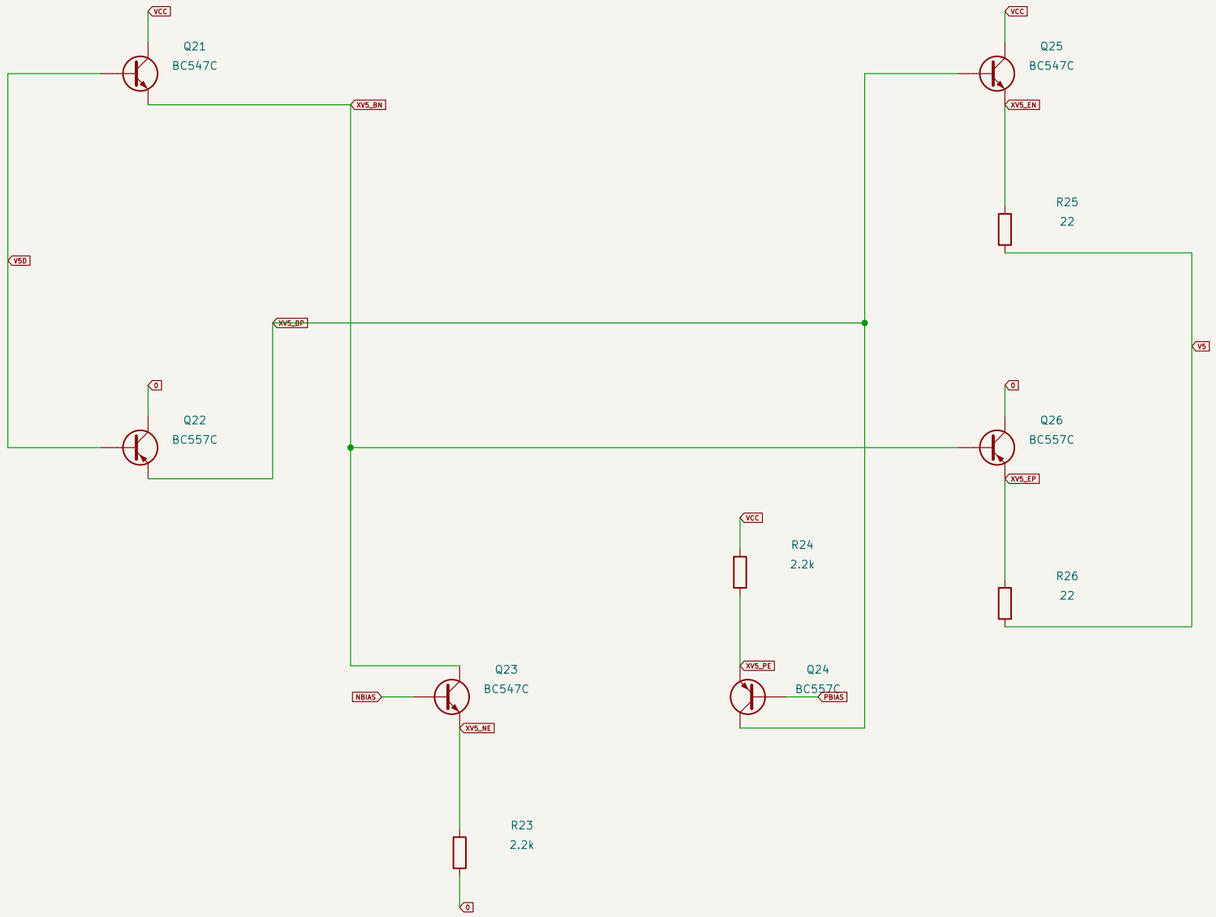


A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /power_cv/
File: power_cv.kicad_sch

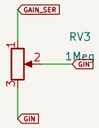
Title: Mamut VCF-1 A0 / power_cv

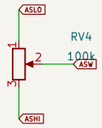
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 2/32

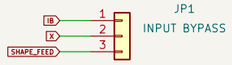


A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /xv5/		
File: xv5.kicad_sch		
Title: Mamut VCF-1 A0 / xv5		
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 3/32





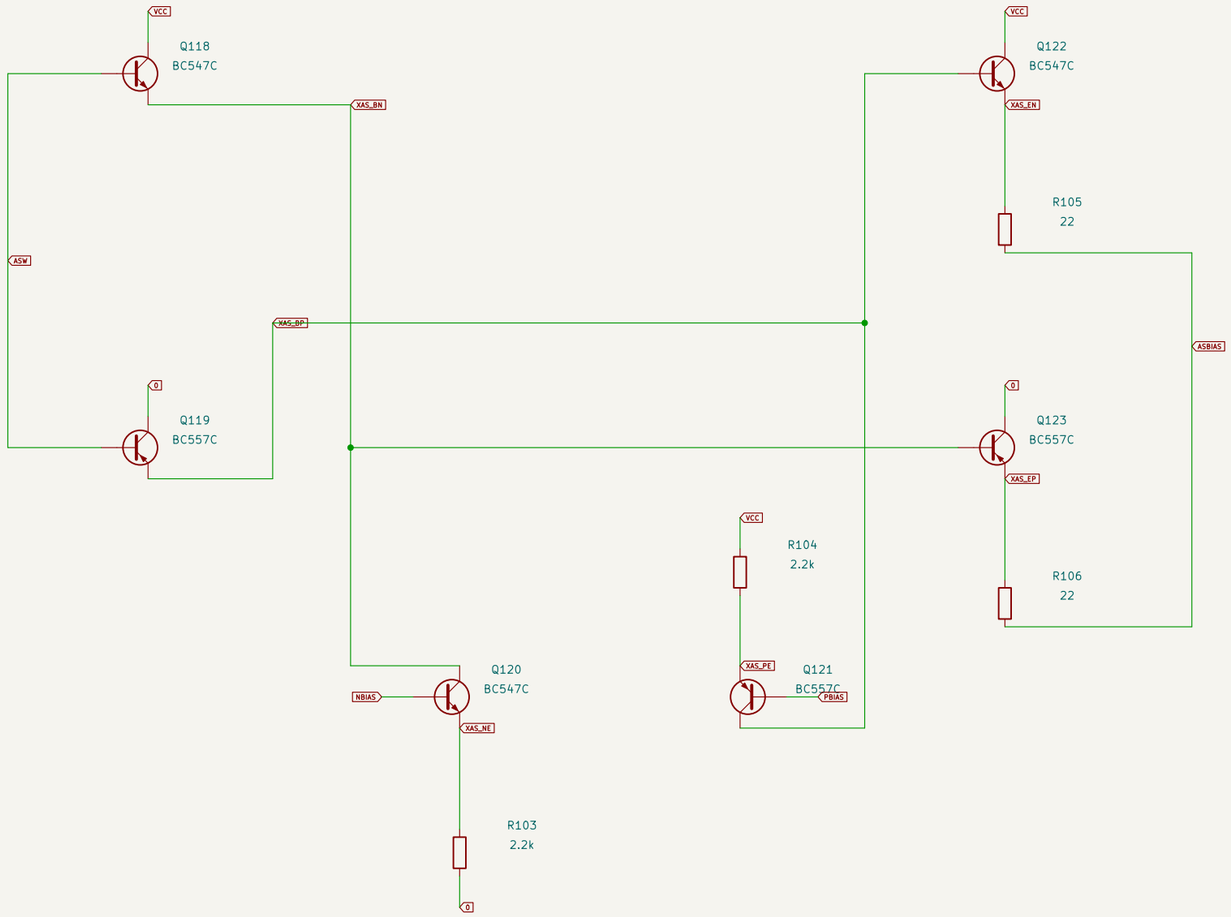


A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /input_shape_2/
File: input_shape_2.kicad_sch

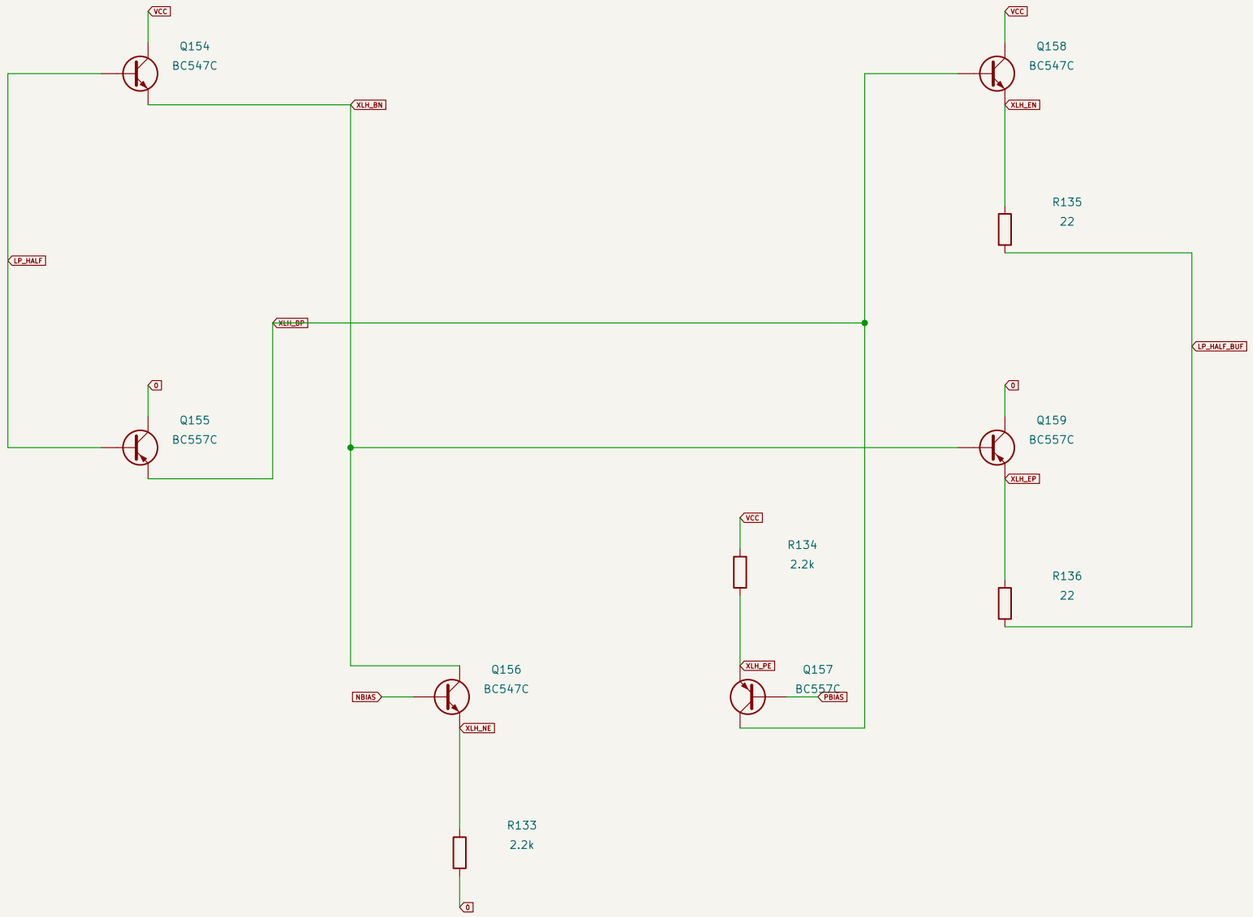
Title: Mamut VCF-1 A0 / input_shape_2

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 4/32



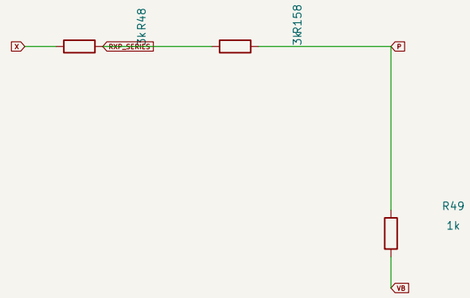
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

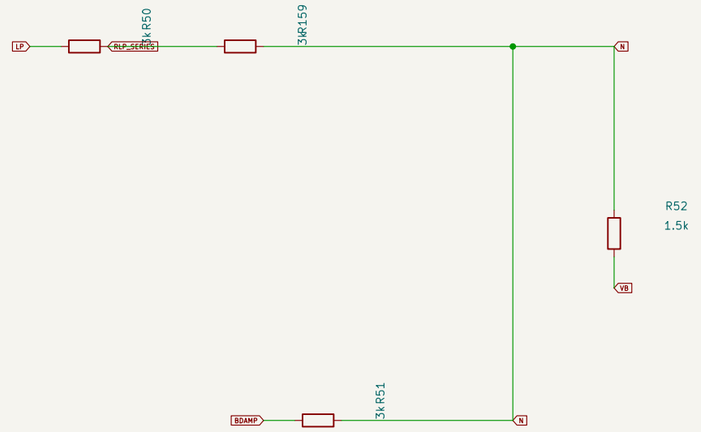
Sheet: /xas/		
File: xas.kicad_sch		
Title: Mamut VCF–1 A0 / xas		
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 5/32

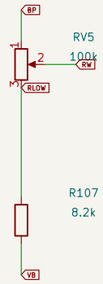


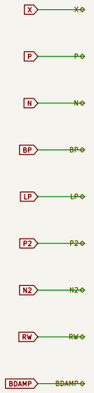
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

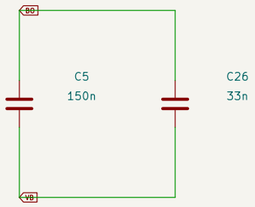
Sheet: /xlh/		
File: xlh.kicad_sch		
Title: Mamut VCF-1 A0 / xlh		
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 6/32

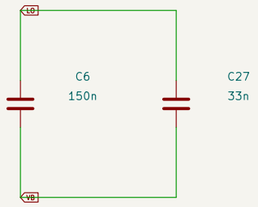


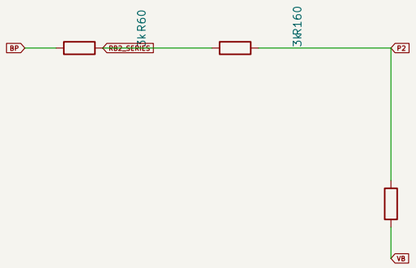












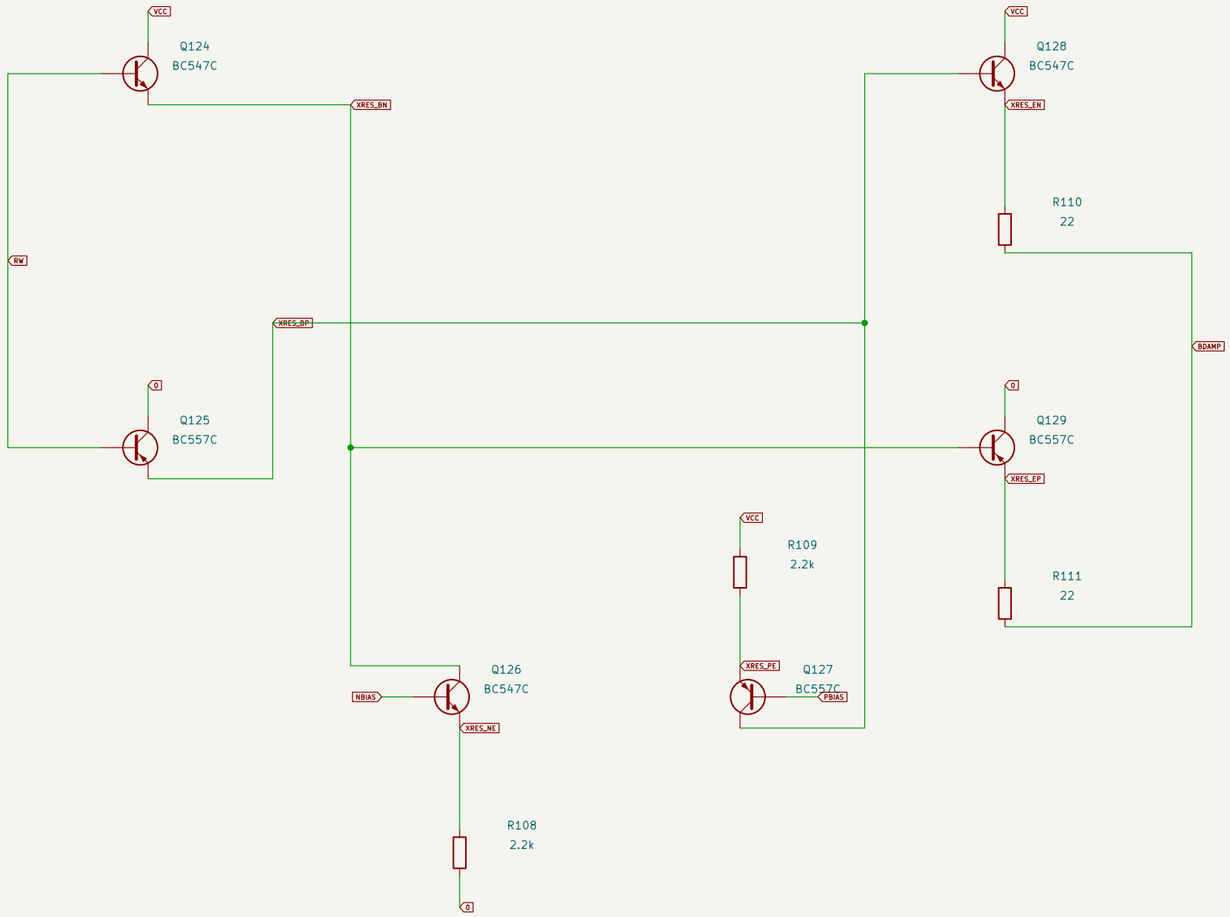


A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /Core summer and timing/
File: core_sum.kicad_sch

Title: Mamut VCF-1 A0 / core_sum

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 7/32





A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /Resonance buffer/
File: xres.kicad_sch

Title: Mamut VCF-1 A0 / xres

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 8/32



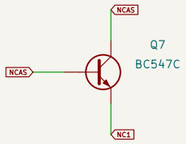
R2
10k



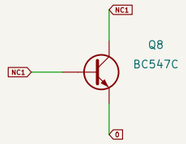
R3
10k



R8
10k



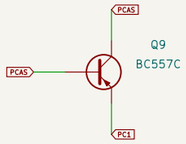
Q7
BC547C



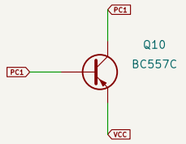
Q8
BC547C



R9
10k



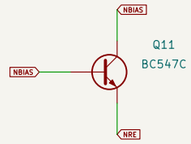
Q9
BC557C



Q10
BC557C



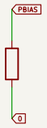
R10
18k



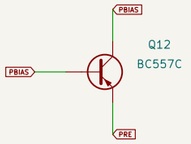
Q11
BC547C



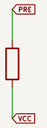
R11
2.2k



R12
18k



Q12
BC557C



R13
2.2k



R21
14k



R22
10k



C7
100u



C8
100n



C9
100u



C10
1u



C11
1u



C12
1u



C13
1u



C14
1u

A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /bias_1/
File: bias_1.kicad_sch

Title: Mamut VCF-1 A0 / bias_1

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 9/32

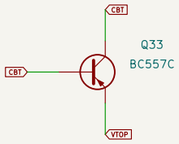


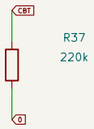


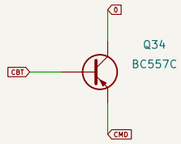


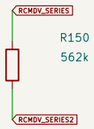












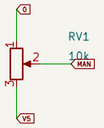


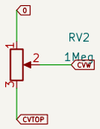










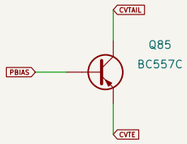


A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

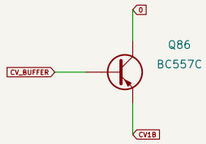
Sheet: /command/
File: command.kicad_sch

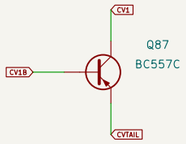
Title: Mamut VCF-1 A0 / command

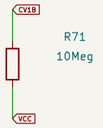
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 10/32

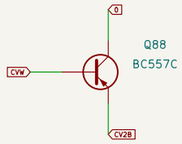


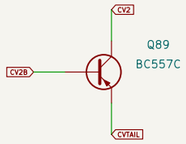


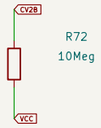


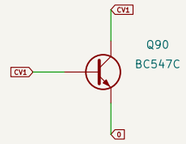


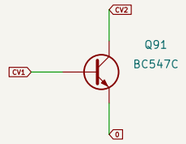




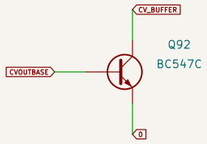
















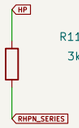
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

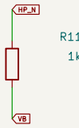
Sheet: /cv_buffer/
File: cv_buffer.kicad_sch

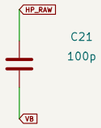
Title: Mamut VCF-1 A0 / cv_buffer

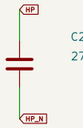
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 11/32

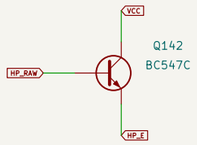


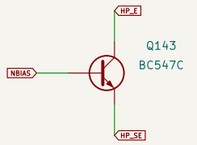




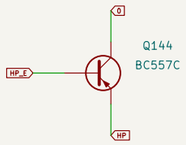


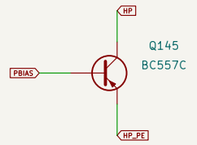




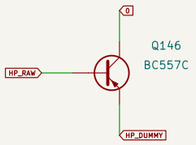


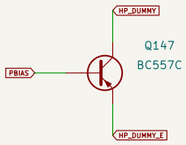




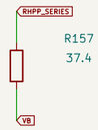


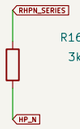










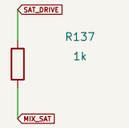


A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /hp_reconstruction/
File: hp_reconstruction.kicad_sch

Title: Mamut VCF-1 A0 / hp_reconstruction

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 12/32







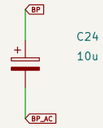




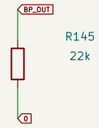


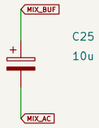






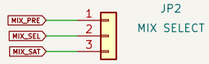












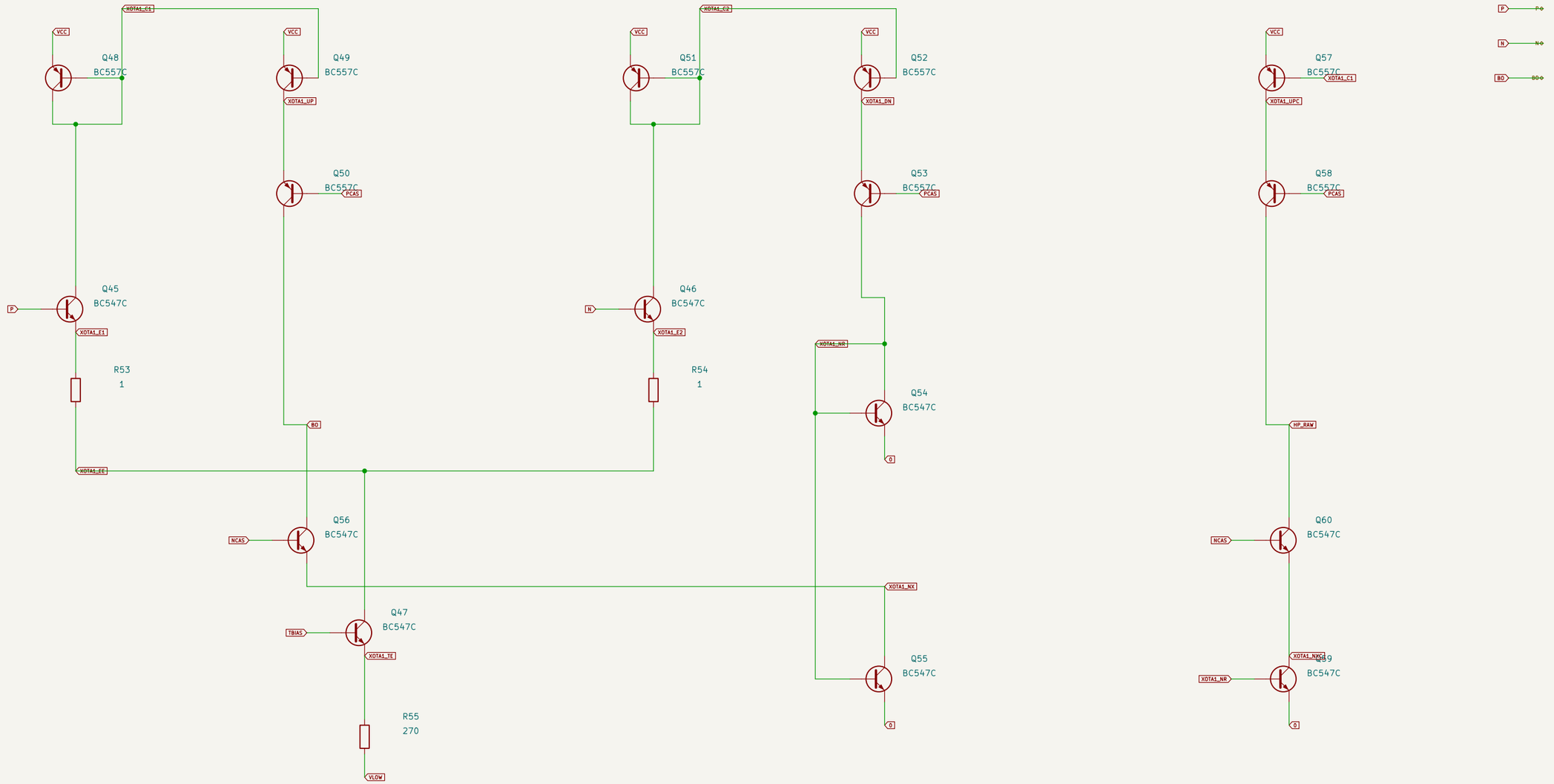


A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /outputs/
File: outputs.kicad_sch

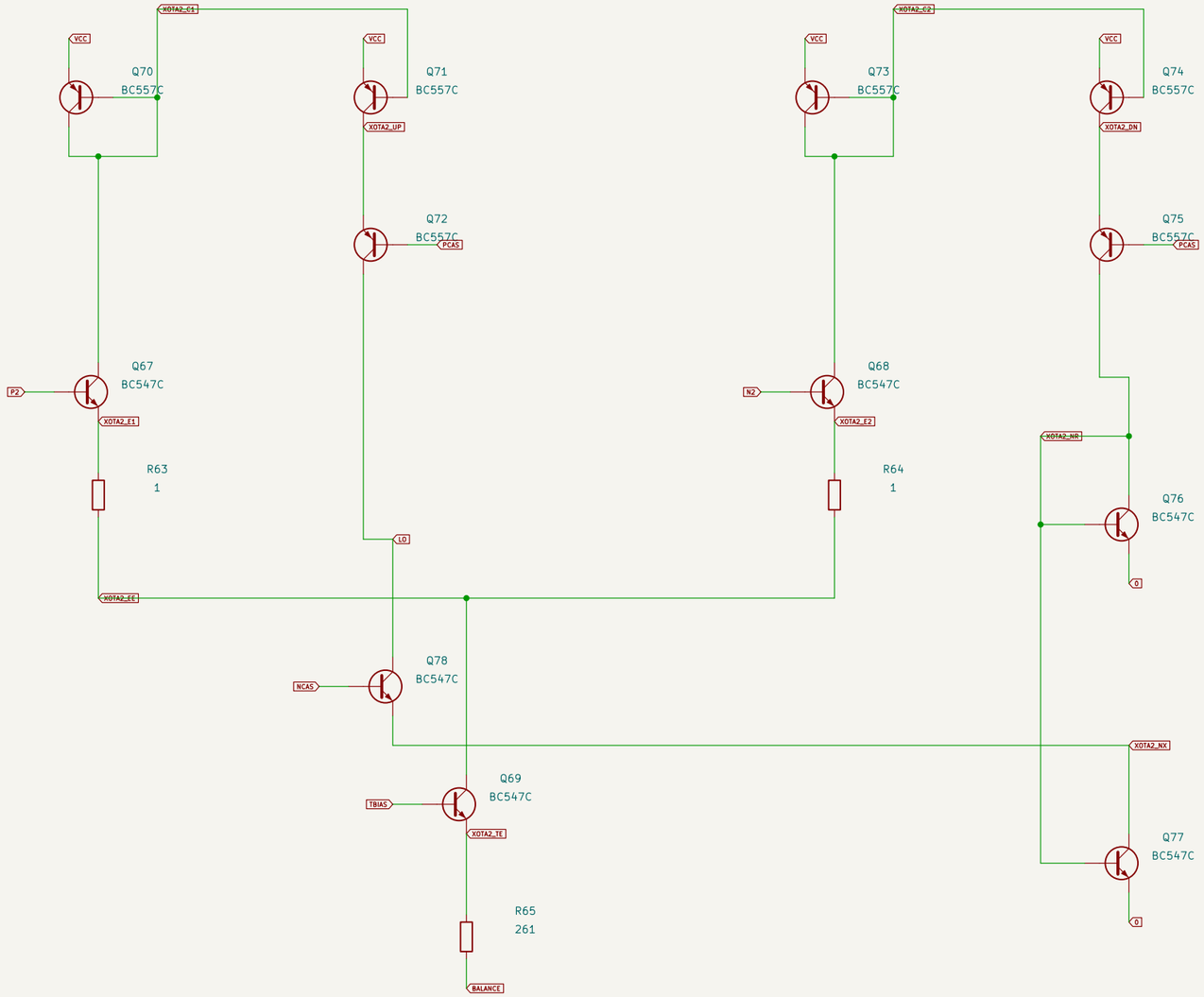
Title: Mamut VCF-1 A0 / outputs

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 13/32



A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /Integrator 1/		
File: xota1.kicad_sch		
Title: Mamut VCF-1 A0 / xota1		
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 14/32

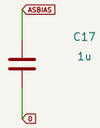


A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

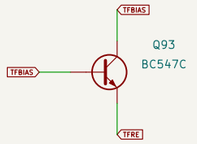
Sheet: /Integrator 2/		
File: xota2.kicad_sch		
Title: Mamut VCF-1 A0 / xota2		
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 15/32

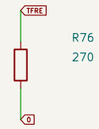










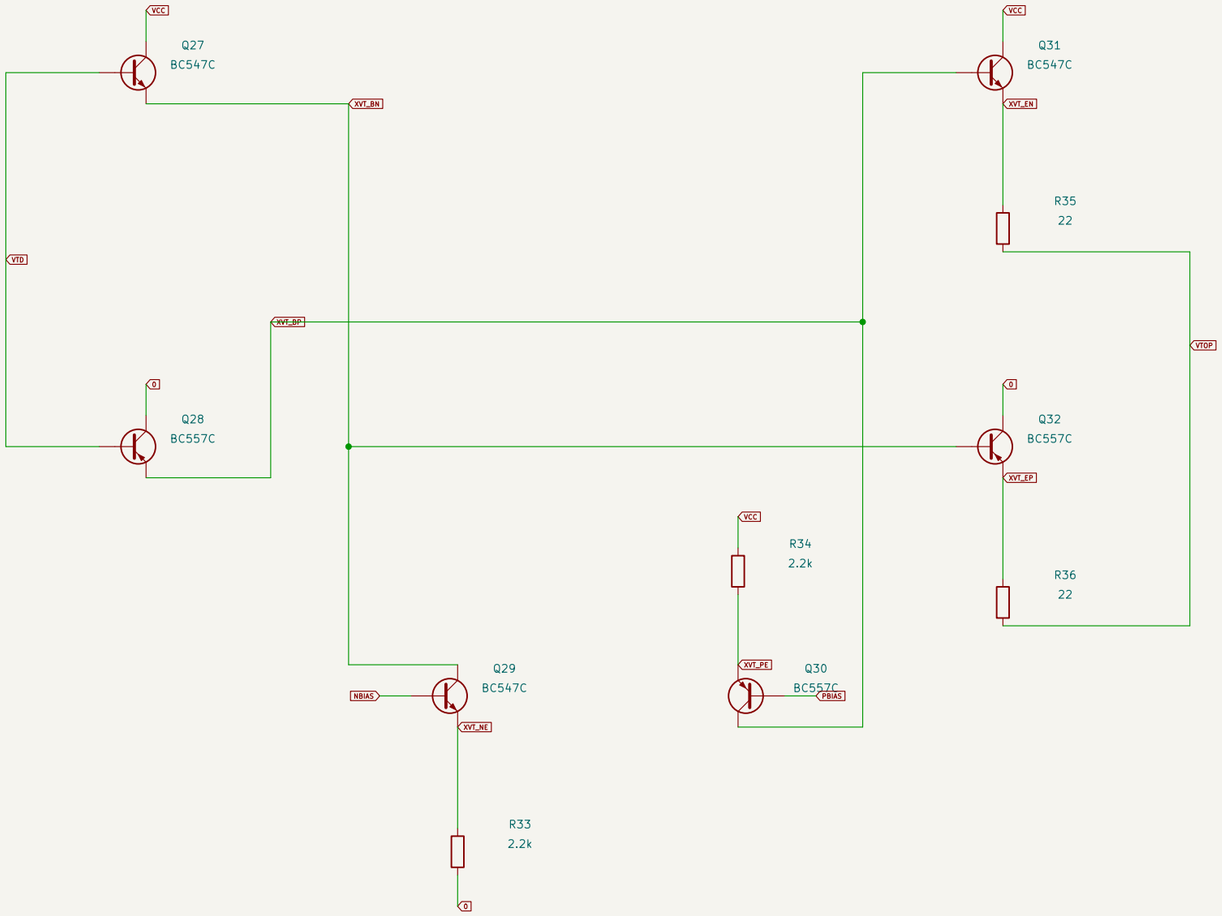


A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /bias_2/
File: bias_2.kicad_sch

Title: Mamut VCF-1 A0 / bias_2

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 16/32

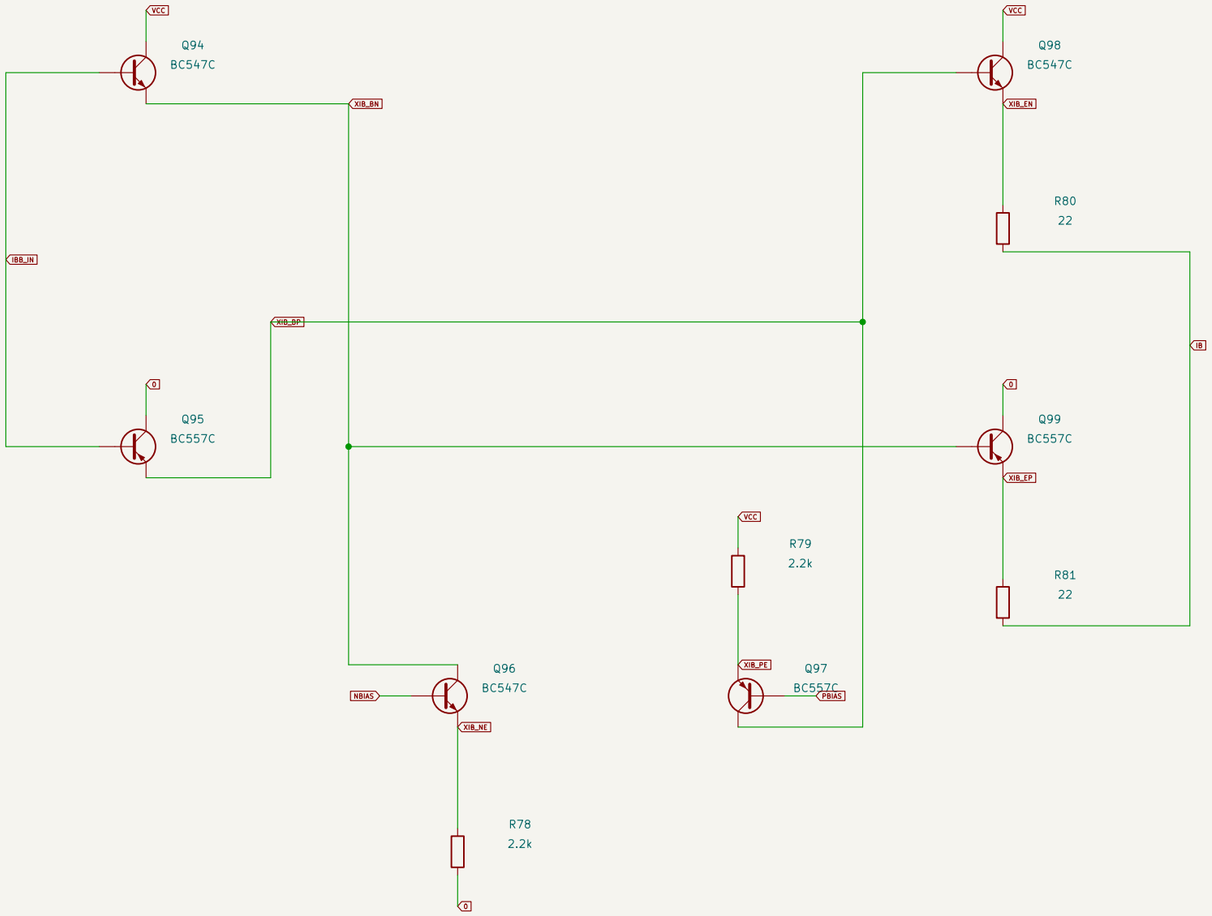


A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /xvt/
File: xvt.kicad_sch

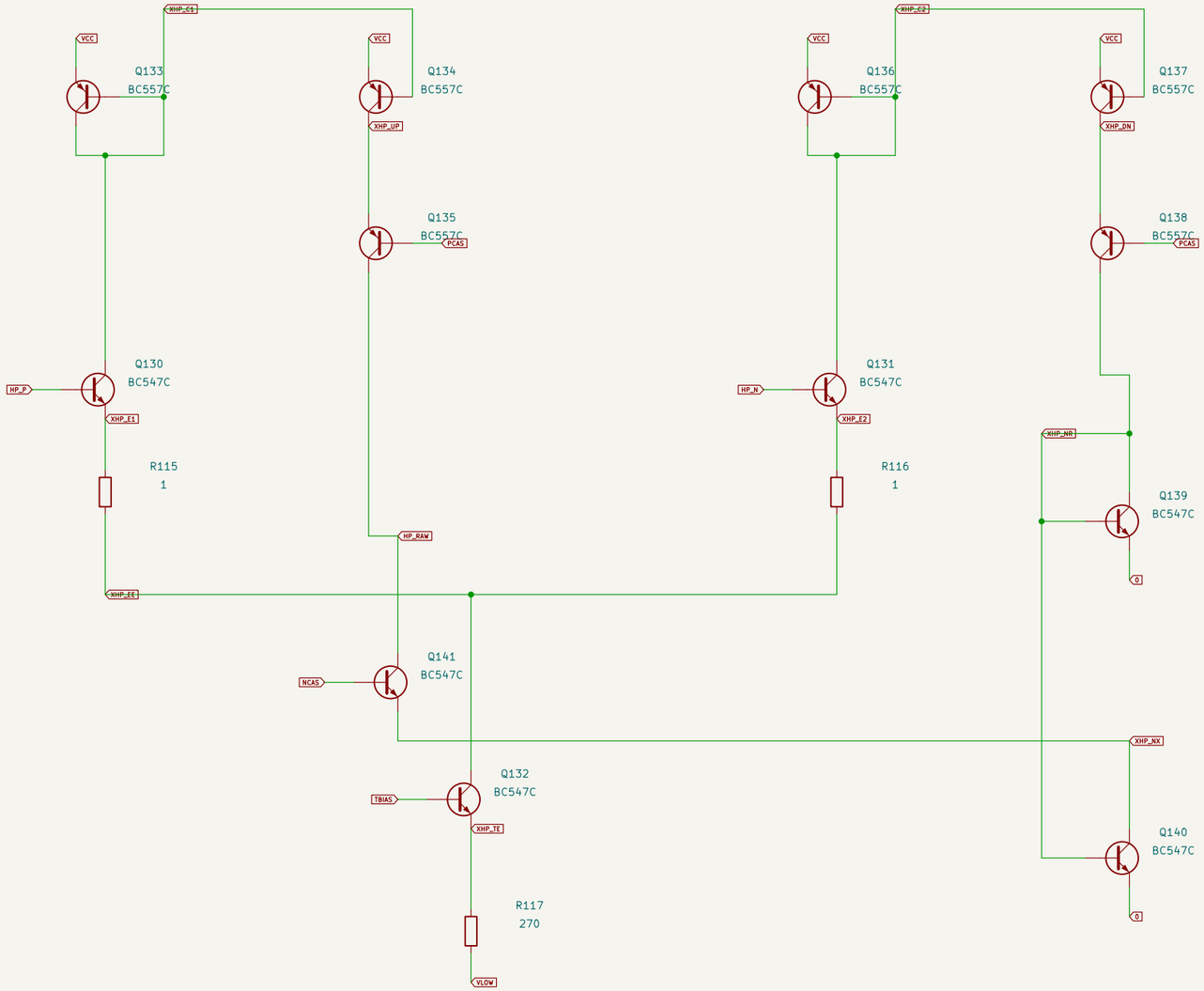
Title: Mamut VCF-1 A0 / xvt

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 17/32



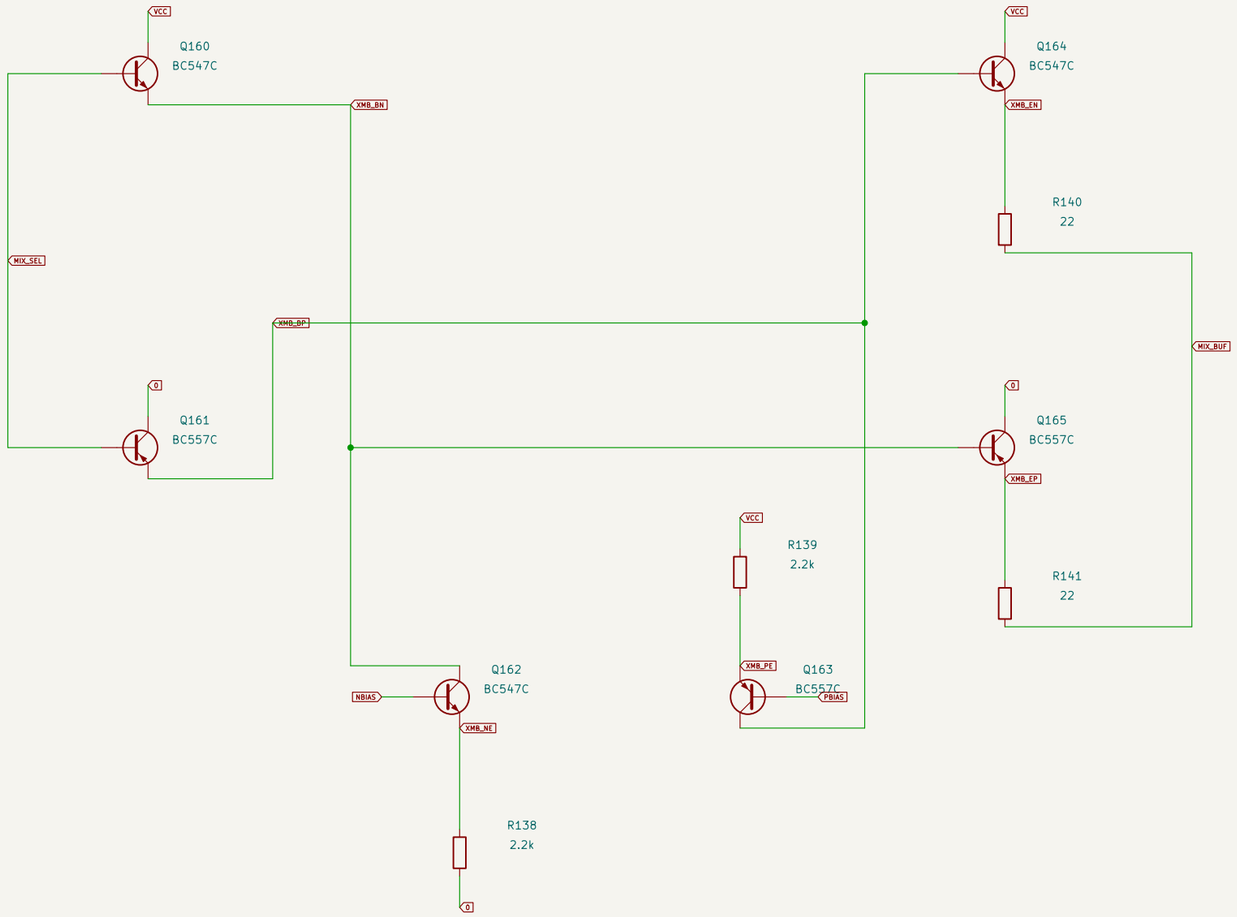
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /xib/		
File: xib.kicad_sch		
Title: Mamut VCF-1 A0 / xib		
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 18/32



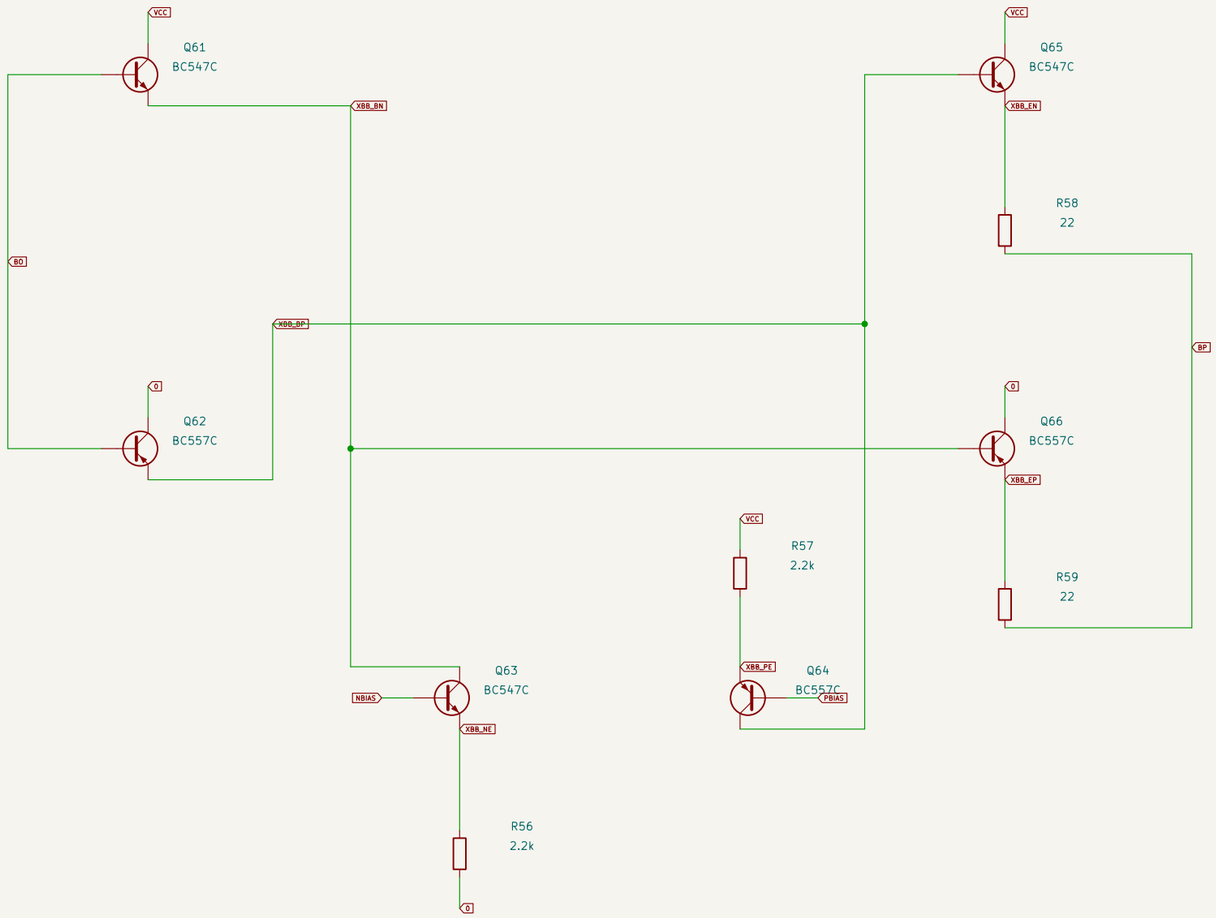
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /xhp/		
File: xhp.kicad_sch		
Title: Mamut VCF-1 A0 / xhp		
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 19/32



A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /xmb/		
File: xmb.kicad_sch		
Title: Mamut VCF-1 A0 / xmb		
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 20/32



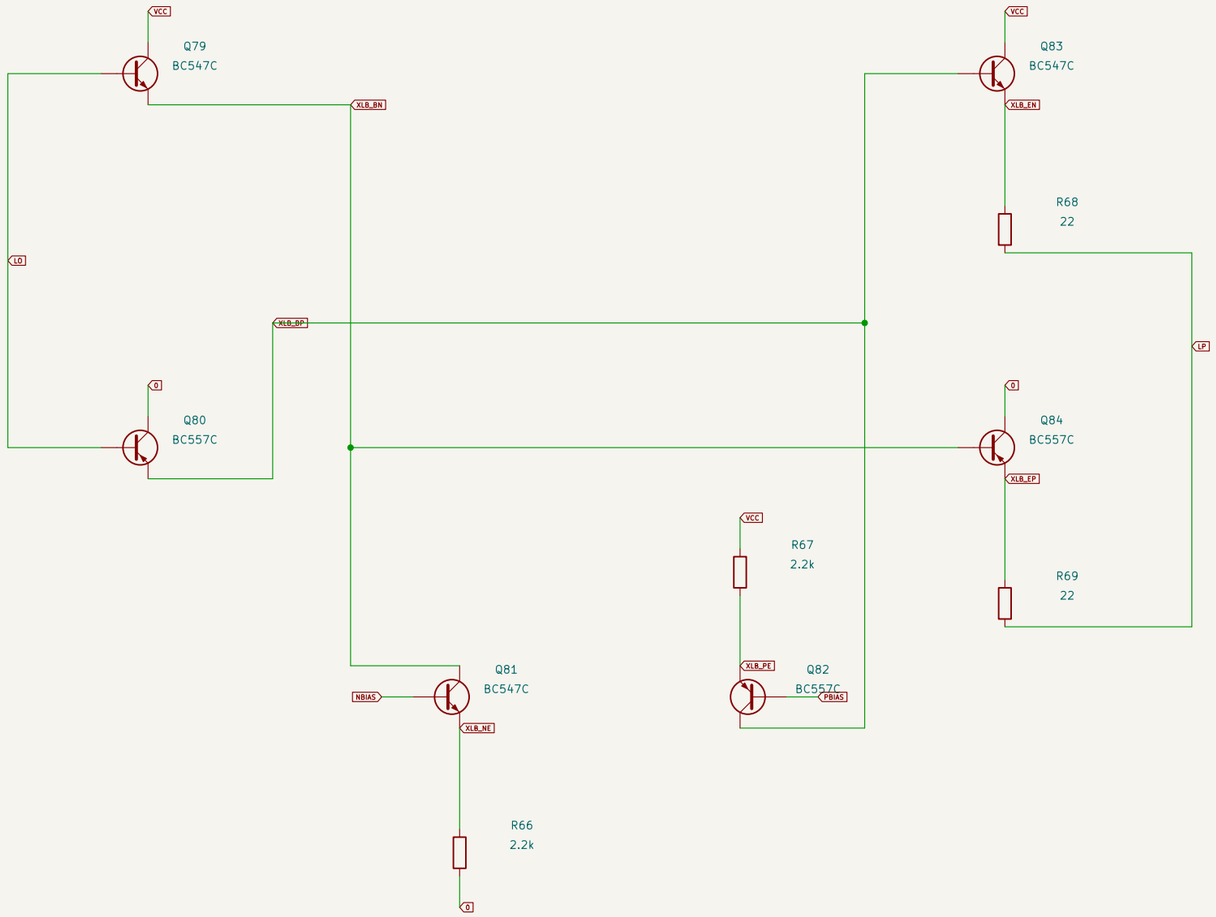


A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /BP buffer/
File: xbb.kicad_sch

Title: Mamut VCF-1 A0 / xbb

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 21/32

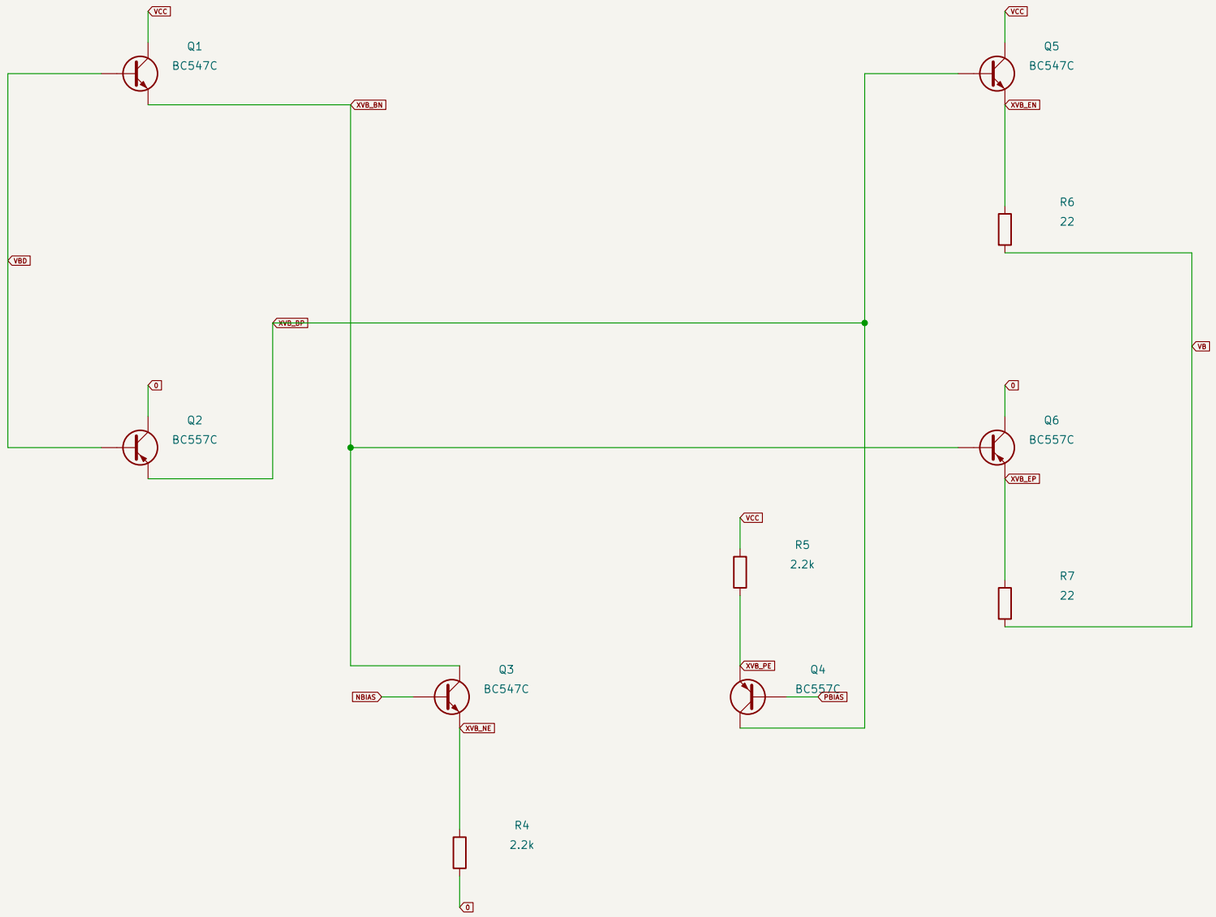


A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /LP buffer/
File: xlb.kicad_sch

Title: Mamut VCF-1 A0 / xlb

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 22/32

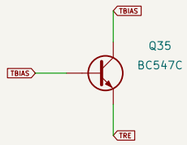


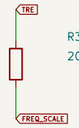
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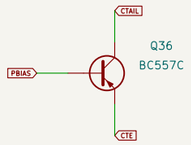
Sheet: /xvb/
File: xvb.kicad_sch

Title: Mamut VCF-1 A0 / xvb

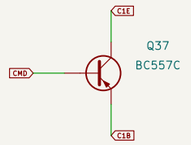
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 23/32

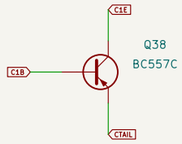


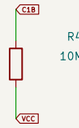


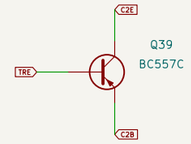


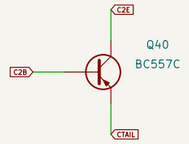


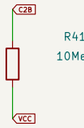


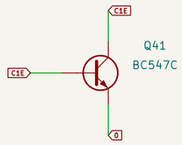


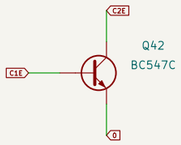


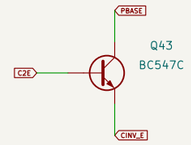








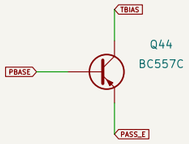






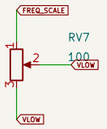










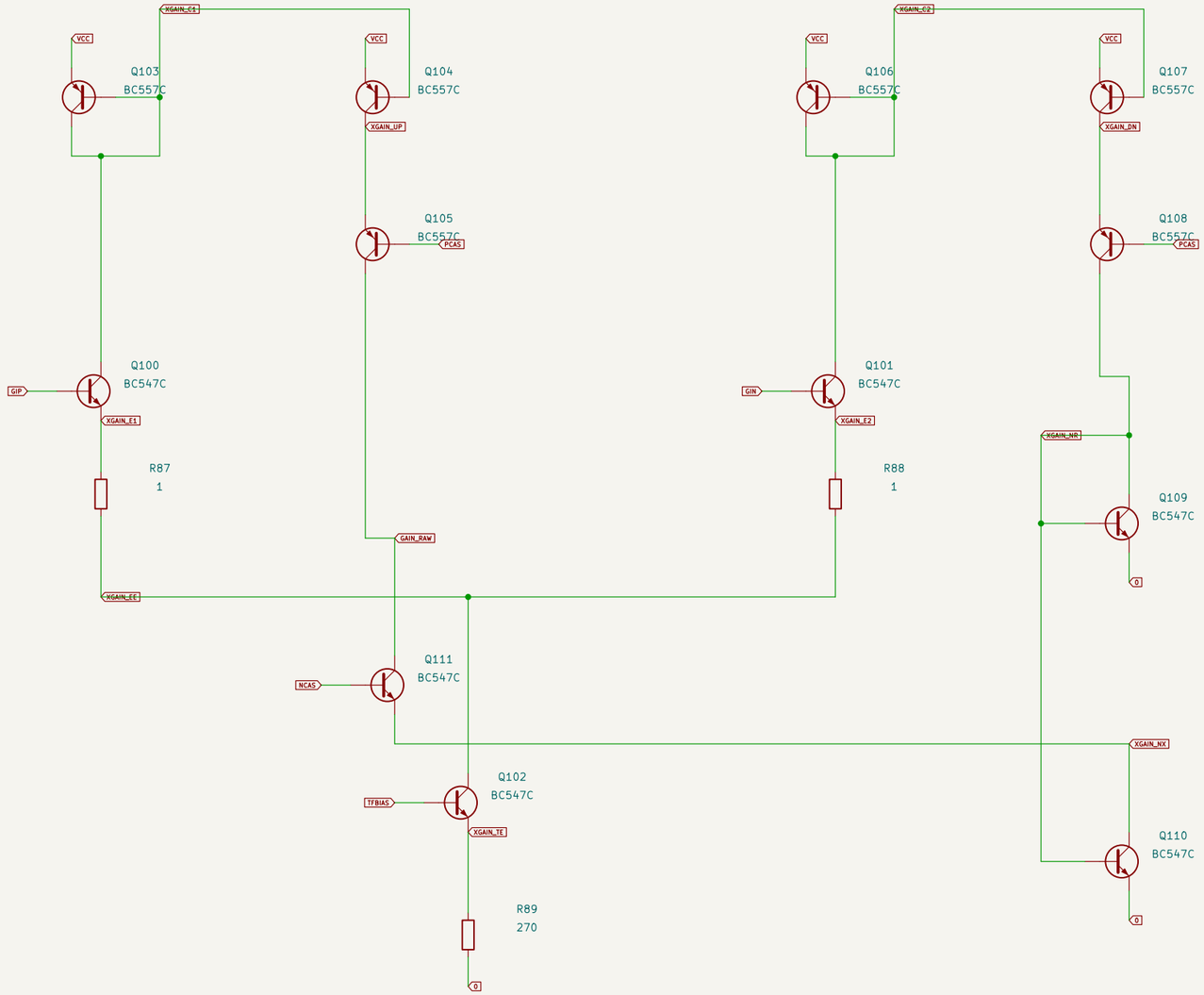


A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

Sheet: /current_servo/
File: current_servo.kicad_sch

Title: Mamut VCF-1 A0 / current_servo

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 24/32

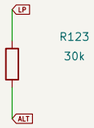


A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

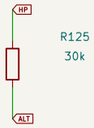
Sheet: /xgain/		
File: xgain.kicad_sch		
Title: Mamut VCF-1 A0 / xgain		
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 25/32

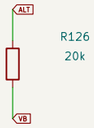


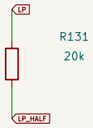




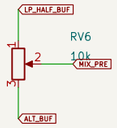










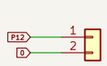


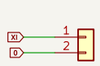
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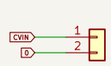
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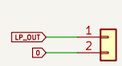
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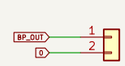
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KiCad E.D.A. 10.0.7		Id: 26/32

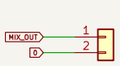










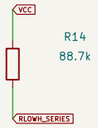


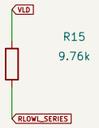
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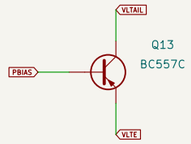
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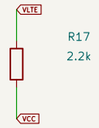
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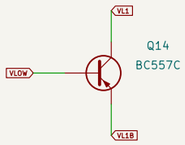


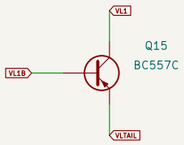


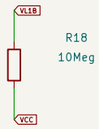


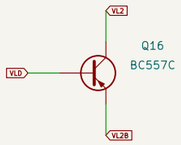


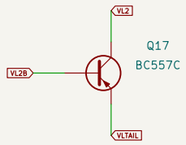


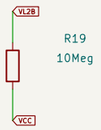


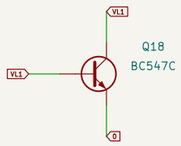


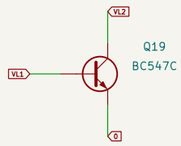


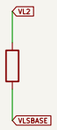


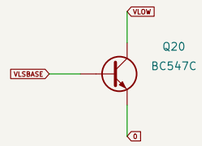


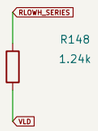


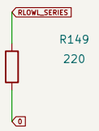










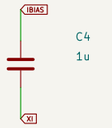


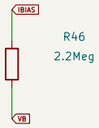
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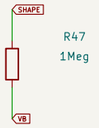
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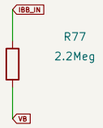
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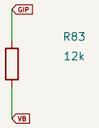


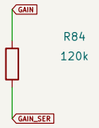


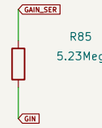


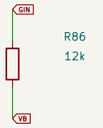


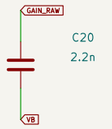


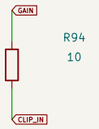


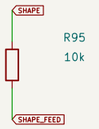


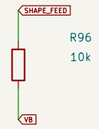


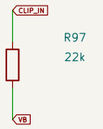






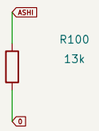


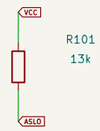
















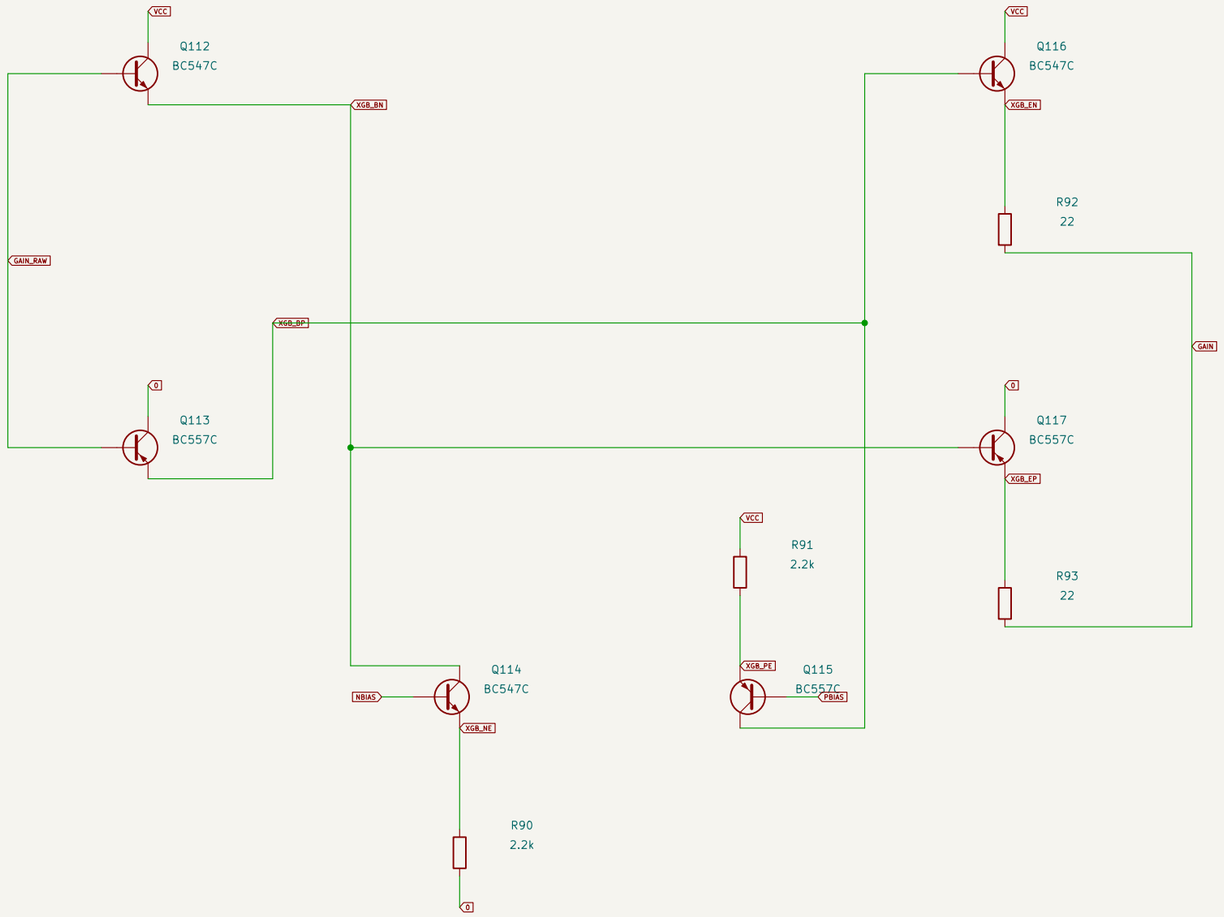






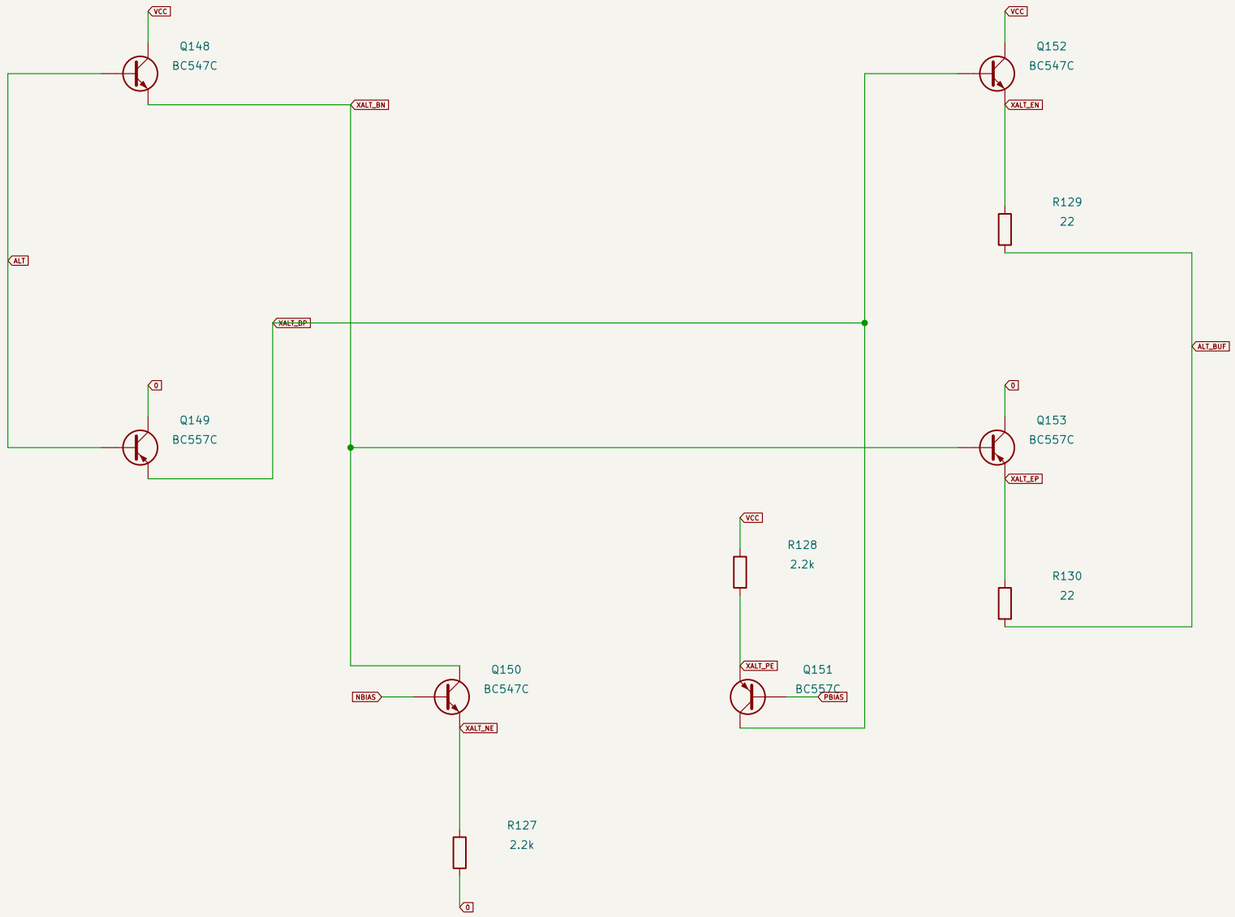
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

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Size: A3	Date:
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A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

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A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

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Size: A3	Date:	**Rev: A0**
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A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VB = INTERNAL BIAS

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